

Shubham Gupta

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Embedded Systems Engineer working on sensor fusion, on-device ML inference, and audio for wearable hardware.

EXPERIENCE

Rivet Industries, Inc

Software Engineer

Palo Alto, US

Oct 2024 – Present

- Led on-device ML infrastructure and optimizations to significantly reduce accelerator latency.
- Led on-device audio algorithms for spatial filters, noise reduction, wake word, and speech to text LLMs.
- Architected real world to display benchmarking suite to quantify 6DoF tracking and display rendering errors.
- Scaled IMU-magnetometer alignment, soft iron, and IMU thermal calibration algorithms from prototype to EVT builds.

M²LLnES, New York University

Research Associate | Advisor: Dr. Laure Zanna

New York City, US

Aug 2024 – Oct 2024

- Eliminated inter-epoch synchronization bottlenecks using Nsight Systems, saving 2 hours/day on NERSC & NYU clusters.
- Optimized GPU residency via asynchronous data prefetching and worker persistence, reducing I/O stalls.
- Implemented Mixed Precision Training, reducing GPU usage by 64% during the training phase while maintaining skill.

NYU ARC Robotics

Computer Vision and DevOps Lead | Mentor: Chris DiMauro

New York City, US

Sep 2022 – Jun 2024

- Secured 5th place in RMNA 2023 internationally against elite robotics teams from USA, Japan and Singapore.
- Led large scale HPC experimentation for training and PTQ tuning int16 quantized object detection models.
- Introduced containerization and led the OTA deployment of controls, sensors, and model inference microservices.
- Established two-way data connections using UART, Ethernet protocol between devices - Nvidia Jetson and Arduino.

HPE Aruba Networking

Software Development Intern

Bangalore, India

Jan 2022 – Jul 2022

- Developed SpringBoot APIs integrating Elasticsearch and Postgres data for edge-to-cloud IoT Operations.
- Designed and deployed a monitoring dashboard for 150k customer device statistics using Elasticsearch, Grafana, and SQL.

Indian Space Research Organization

Research Intern | Advisor: Dr. Hebbar R

Bangalore, India

Aug 2019 – Apr 2020

- Managed data annotations and experiments, published findings as lead author at the CVMI 2022 conference.
- Built an automated Sentinel-2 data pipeline with DeepLabv3+ and SnaPy/GDAL for water body segmentations.

EDUCATION

New York University

Master of Science in Computer Engineering — 3.8/4.00

New York City, US

PES University

Bachelor of Technology in Computer Science and Engineering — 8.39/10.0

Bangalore, India

PUBLICATIONS

Dheeshjith, S., Subel, A., **Gupta, S.**, Adcroft, A., Fernandez-Granda, C., Busecke, J., & Zanna, L. (2024). Transfer Learning for Emulating Ocean Climate Variability across CO₂ forcing. arXiv preprint arXiv:2405.18585. [\[preprint\]](#) [\[ICML Oral 2024\]](#)

Lee, K., **Gupta, S.**, Kim, S., Makwana, B., Chen, C., & Feng, C. (2023). SO-NeRF: Active View Planning for NeRF using Surrogate Objectives. arXiv preprint arXiv:2312.03266. [\[preprint\]](#) [\[code\]](#)

Gupta, S., Uma, D., & Hebbar, R. (2023). Analysis and application of multispectral data for water segmentation using machine learning. In Computer Vision and Machine Intelligence: Proceedings of CVMI 2022 (pp. 709-718). Singapore: Springer Nature Singapore. [\[preprint\]](#) [\[code\]](#) [\[publication\]](#) [\[slides\]](#)

PROJECTS

Cavemen: A prehistoric approach for Mapless Navigation | [\[Website\]](#) | *Python3, SIFT, FAISS, VBoW, Redis*

2023

- Visual Place Recognition and navigation using only monocular input and classic visual features. **Top submission** for coursework Robot Perception ROB-GY 6203 Fall 2023. [\[LinkedIn\]](#)

LYCB: Leave Your Clothes Behind | [\[Website\]](#) | *HPC, Segment-Anything, COLMAP, Pytorch3D*

2023

- Proof-of-Concept to capture real world clothes on hand held devices and create assets to be used in Blender or virtual try-ons.

TECHNICAL SKILLS

Languages: Python3, C++, Java, L^AT_EX

Libraries: ONNX, TensorRT, QAI²T, FastAPI, ADB, Hydra, Slurm

Applications: NVIDIA Nsight, Docker, Elasticsearch, SQL, Grafana